

Decision-Level Divergence of Independent Coding Agents

A pre-registered measurement of where agents disagree — across arms and across models — with an instrument that had to survive its own failure

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ABSTRACT

Multi-agent coding tools rest on an unmeasured premise: that independent agents given the same task produce usefully different solutions rather than N spellings of one solution. We measured it. Fifteen agents re-implemented three completed tickets from a production Rust codebase, from pre-fix states in which the shipped fix was unreachable *by construction*; nine arms shared one model and configuration, and six varied the model. Divergence was scored at the *declaration* layer — what each arm says it decided — never the diff, under rules committed to version control before any result existed. Three findings. First, divergence is *region-shaped*: arms converge wherever the environment determines the design (all fifteen arms chose the same system interface for the same feature) and fork wherever the ticket underdetermines it (three ranking policies inside an identical two-file envelope; five replay-suppression designs across eight arms). Second, file-grain metrics lie in both directions — our pre-registered surface metric scored a genuine three-way policy fork as 0.000 divergence and changelog bookkeeping as 0.583 — and a claim-level repair, validated retroactively against cases whose truth was known, inverted the ordering correctly. Third, varying the model within one family produced mechanism-level designs that three same-configuration arms never produced (verdict R1 under the pre-registered rule, with a sensitivity analysis that we publish rather than smooth away), at materially different token prices for identical briefs. The practical consequence for tooling: fork *decisions*, not tasks; compare declarations, not diffs; let humans, not models, judge whether two decisions are the same. The method ships as an installable skill in the repository accompanying this report.

1 Introduction

A team that dispatches several agents against one ticket is making a bet: that the agents will disagree in ways worth paying for. If they converge, the elimination and comparison machinery downstream has nothing to work on, and the N-fold token bill purchased one opinion. If they diverge, the follow-up questions decide the tooling’s whole shape: *where* do they diverge, can the divergence be located cheaply, and does adding model diversity buy disagreement that same-configuration sampling cannot?

These questions are empirical, and to our knowledge nobody had answered them at the level that matters — the decisions agents make, rather than the text they emit. Two spellings of one idea produce different diffs; a diff-level or file-level comparison therefore measures prose variance, not judgement variance. We built and validated an instrument for the decision level, ran two pre-registered studies against it, and report everything, including the failure of our own first metric and the sensitivity of our headline verdict to two flagged adjudication calls.

This report accompanies a working method — the *Crossfire* protocol, released as an installable agent skill — and underpins the Crossfire feature of Conclave, our agent-development product for enterprise teams. The research is published so that the claims the tooling makes can be checked against the measurements that

produced them, and so that progress on the open questions in Section 7 is trackable in public.

2 Related work

Effective false positives. Sadowski et al. describe Google’s operating rule for analysis surfaces at code review: below 10% *effective* false positives — defined behaviourally, as findings after which “developers did not take some positive action” — or the surface is dismissed and disabled [1]. We adopt both the definition and the discipline: our instrument’s own first version failed behaviourally and was repaired under pre-registered criteria.

Self-report is inverted. METR’s randomised trial found experienced developers were 19% slower with AI assistance while believing themselves 20% faster [2]. This is why no step of our pipeline asks a model (or a person) whether two decisions are “really the same” as a free-form judgement: sameness is decided by a mechanical rubric with quoted evidence, or by a human against that rubric, never by an unstructured model call.

Coordination beats merging. The AgentRoom preprint measured concurrent multi-agent coding under a file-claim protocol and found that pooling independent outputs — parallel-merge — scored *below a single agent working alone*, while coordinated agents scored above [3]. Our protocol therefore never text-merges arms: comparison happens at the declaration layer and recomb-

nation happens by a written brief that a fresh run implements once.

3 Method

3.1 Corpus and tickets

Three completed tickets were selected from the history of a production GPU-accelerated terminal emulator (Rust, roughly 41 kLOC, several hundred commits), against written criteria: a single non-merge commit; 60–600 changed lines; a commit message whose *problem* statement is separable from its *approach*; and genuine design latitude. One candidate from a second codebase was rejected for size (32 changed lines), recorded as a limitation rather than silently swapped. The tickets: a reveal-in-file-manager feature (**t1**), a session-adoption ranking bug (**t2**), and a false-completion notification bug (**t3**).

3.2 Contamination control by construction

Each arm received a fresh workspace built as `git init; git fetch <repo> <base-sha>; checkout FETCH_HEAD`. Fetching a commit brings its ancestry only; the shipped fix is a *descendant* of the base and therefore does not exist in the arm’s universe — absence by construction, not by instruction. Every workspace was verified before launch: tip equals base, and reachable objects equal ancestry count. Arms were further instructed not to read other copies of the project, use the network, or spawn sub-agents, and these weaker instruction-level controls are recorded as such.

The ticket briefs were written from the problem halves of the real commit messages by an author who had read the real fixes; the briefs are published verbatim in the repository precisely so they can be audited for approach leakage, and the one place a brief most plausibly steered the arms is named in Section 5.

3.3 The declaration layer

Each arm ships, beside its diff, a structured declaration: **decided** entries `{id, surface, claim}` — one line per significant decision, anchored to a file path or named concern; **least_confident** references into **decided**; and **not_done**. Arms did not compile or run anything (a uniform scope choice that keeps arms cheap and comparable) and were required to declare unverified surface rather than fake verification.

3.4 Pre-registration by commit ordering

Every rule — ticket selection criteria, metrics, exclusion and repair policy, decision thresholds, and later the repaired instrument and the model-axis criteria — was committed to version control *before* the data it governs existed. The proof of ordering is the git history itself: a commit containing rules and no results, followed by commits containing results and no new rules. Four such boundaries exist across the two studies.

3.5 Metrics

The pre-registered primary metric was surface divergence: mean pairwise Jaccard distance over declared surfaces,

$$D(A, B) = 1 - \frac{|S_A \cap S_B|}{|S_A \cup S_B|}, \quad (1)$$

averaged over arm pairs, with contested-surface share and confidence-consensus alongside, and a declaration-independent secondary computed from the diffs’ file sets. The decision rule, fixed in advance: divergence below 0.15 with contested share above 0.80 on all tickets reads *converge*; above 0.40 on all tickets reads *diverge*; anything mixed is reported as mixed.

3.6 The claim-level instrument

Because Section 4.1 shows the surface metric failing, a claim-level layer was defined and pre-registered between studies. Claims from different arms answering the same design question form a *cluster*; each cluster carries one topic from a fixed taxonomy (mechanism, policy, threshold, compat, presentation, verification) and one verdict:

- *fork* — the claims commit to materially different behaviour: a different interface, signal, or control structure, or the same shape with a constant differing by at least $2\times$;
- *same* — differences are wording or incidental detail;
- *complementary* — disjoint, non-conflicting aspects.

Fork share is forks over clusters with at least two arms,

$$F = \frac{\text{fork clusters}}{\text{clusters with } \geq 2 \text{ arms}}. \quad (2)$$

Every verdict is recorded with a quoted justification so that a disagreeing reader can re-run the call. The retroactive validation criteria, fixed before any tag existed: the amended metric must read the known hidden fork of **t2** as divergence (P1), and must order **t2** above **t1**, inverting the file-grain ordering (P2). Failure of either would have halted the programme.

4 Results

4.1 Study 1: nine same-configuration arms

Nine arms (three per ticket; one frontier model, identical briefs, shared base commits, fresh context each) produced nine valid declarations — no exclusions, no repairs. Mean cost 143,197 tokens per arm (range 126k–172k), 1.29M total, within 5% of the estimate.

Ticket	Surface div.	Contested	Rule side
t1-reveal	0.583	50%	diverge
t2-adoption	0.000	100%	converge
t3-bell	0.333	50%	between

Table 1: Pre-registered surface metric, study 1. Three tickets, three sides: the stopping rule required unanimity and did not fire.

The instructive result is not the mixed verdict but the metric’s double failure. **t2-adoption** scored 0.000 — textbook convergence — while its declarations record three different ranking policies (a 72-hour save-time handicap; a hard substance tier inside a 7-day window; a 7-day handicap) inside an identical two-file envelope. **t1-reveal** scored 0.583 — the “diverge” side — while all three arms independently selected the same system interface, threading model, authority rule, and fallback; the score was driven by whether an arm listed the changelog among its surfaces, and the

file-set secondary reads t1 at 0.167. A file-grain metric measures what arms choose to *declare*, not what they *decided*.

4.2 Study 1b: the repaired instrument, validated on known truth

The claim-level layer clustered the nine declarations into 28 design questions. Both pre-registered criteria passed:

Ticket	Clusters	Fork share	Where the forks are
t1-reveal	12	0.167	decode semantics only
t2-adoption	9	0.222	ranking policy; its constant
t3-bell	7	0.286	claim lifetime; re-play signal

Table 2: Claim-level re-scoring of study 1. P1: the hidden t2 fork registers (0.222 vs the file-grain 0.000). P2: the ordering t2 > t1 inverts the file-grain ordering.

Confidence consensus moved from one degenerate point per ticket (always the ticket’s main file) to ten cluster-grain points, *four of which sit exactly on fork clusters*: arms were least confident precisely where they disagreed. The declaration layer does not merely measure forks; it locates them.

4.3 Study 2: the model axis

Six further arms — one Opus and one Sonnet per ticket, identical briefs and bases, blind to study 1 — were scored against study 1’s clusters, each arm’s stance per cluster being *aligned*, *fork*, or *unaddressed* relative to the study-1 modal design. The pre-registered criterial quantity: forks on the 13 clusters that were mechanism-topic and unanimous across the study-1 arms. Rule: at least 2 such flips, or any cluster where both probe models agree with each other *against* the study-1 consensus, reads R1 (model heterogeneity is load-bearing); otherwise R2.

Result: 3 flips; zero agreement-against-consensus clusters; verdict R1. The flips: a bounded 4 KiB head-read replacing a whole-file read (t2, Opus); a stateless gate-and-drop bell where all three study-1 arms built defer-and-resolve (t3, Sonnet); and, in consequence of the same design, the absence of the 300 ms idle hold every study-1 arm kept (t3, Sonnet).

Sensitivity, published rather than smoothed: the first flip is borderline (the rubric’s letter says fork; a stricter reader says optimisation), and the two Sonnet flips share one root. Under the rule as registered the verdict is R1 at 3 flips; de-duplicating the correlated pair leaves R1 at the threshold; de-duplicating and re-scoring the borderline call yields R2. We do not invent a de-duplication clause after seeing that it would matter — that is the move pre-registration exists to prevent — but the verdict travels with this table, and the single call that grades it is published for audit.

Beyond the criterial count, the probe widened exactly the regions study 1 called underdetermined: replay suppression ended with five distinct designs across eight arms (one probe arm’s declaration explicitly argues against the design the other probe arm

chose), while every environment-determined cluster stayed converged across all three models.

Model	Mean tokens/arm	Arms
Fable (study 1)	143,197	9
Opus	154,286	3
Sonnet	199,386	3

Table 3: Cost of identical briefs by model. The 39% Sonnet premium makes per-model accounting a first-class requirement of any dispatching tool.

5 Threats to validity

Sample. Three tickets, one codebase, one model family. Nothing here generalises beyond that without replication; the roadmap (Section 7) exists because of this row.

Adjudicator position. The clustering and verdicts were made by the same author who wrote the briefs, had read the real fixes, and had seen the interim numbers. Mitigations: criteria and rubric committed before tagging; every verdict quotes its claims; the two calls that decide anything are re-runnable mechanically. The residual bias is a limitation, not a solved problem.

Brief leakage. The t2 brief’s requirement that a stale session “must not become permanently sticky” is the place brief wording most plausibly steered arms — all five arms across both studies engineered a bound the shipped fix lacks. The briefs are published so this can be audited.

No builds. Arms verified by reading; whether compile-fix loops would tighten or loosen convergence is unknown, and the cost figures exclude them.

Within-family models only. Study 2 is a lower bound on heterogeneity; cross-family measurement is open.

6 The method as a practice

The accompanying repository packages the protocol as an installable agent skill: ancestry-only workspace construction; the declaration schema; the clustering rubric and verdict grammar; the scoring tools (which count and never judge); and the pre-registration-by-commit-ordering discipline. The operational shape that the findings recommend to any multi-agent tool:

1. *Fork decisions, not tasks.* Locate underdetermined regions (cheap declaration passes plus confidence consensus find them) and spend arms there; the determined majority of a ticket re-buys the same answer N times.
2. *Compare declarations, not diffs* — at claim grain, with a fixed topic taxonomy; file grain lies in both directions.
3. *Let humans judge sameness* against a mechanical rubric with quoted evidence; never a free-form model call.
4. *Never text-merge arms*; recombine by brief, implement once.
5. *Price arms per model*, and publish verdicts with their sensitivity.

7 Roadmap

Open, tracked publicly in the repository: cross-family model axes; replication on a second codebase (a TypeScript application corpus is prepared); whether cheap declaration-only passes predict where full arms fork (the load-bearing question for region-scoped dispatch); and an interruption-surface study governed by the effective-false-positive definition of ref. [1]. Updates land as dated entries with the same rules-before-results discipline used here.

References

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- [2] METR. “Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity.” 2025.
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- [4] R. Parasuraman, V. Riley. “Humans and Automation: Use, Misuse, Disuse, Abuse.” *Human Factors* 39(2), 1997.
- [5] Brown Family Sports Inc. Study artifacts: pre-registered methods, briefs, declarations, adjudications with quoted verdicts, scoring tools, and per-arm costs. Published in the repository accompanying this report.